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WHITE PAPER · PORTFOLIO & AI GOVERNANCE

# AI Should Make People Better Thinkers, Not Just Faster Producers

What student trials suggest for expert work, and what no large workplace trial has tested

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## EXECUTIVE SUMMARY

**For work that depends on judgment, make the person state a view, the key assumptions, and the evidence that could change it before AI challenges the analysis. The evidence is from students; no large workplace trial has tested it.**

AI-assisted work can arrive faster while the judgment behind it becomes harder to see. This paper takes the position that people should form and defend a view before AI answers. Two student trials, a small experiment with developers, and workplace warning signs show that how AI is used changes what people keep, but no large workplace trial has tested it on experienced people, so the paper proposes a one-quarter pilot.

- 01** In one student trial, an AI tool that gave direct answers left students 17 percent below a no-AI group on a later exam. A guarded version built on the same model, which led students through hints, largely avoided that loss.
- 02** For high-judgment work, ask the person to write down a first view, the assumptions it needs, and the evidence that would change their mind before AI responds. The paper calls this a short pre-mortem. A reviewer who knows the subject then checks that the view came first, the assumptions could be wrong, and each challenge was answered.
- 03** Pilot the approach for one quarter in one function with a named executive owner, and compare speed, rework, and errors with an earlier quarter. That shows cost and fit, not proof, and the pilot should stop if neither rework nor errors fall.

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## When faster output hides weaker judgment

Suppose your team now turns in AI-assisted work faster. One recommendation reads well and reaches review early. A leader asks which assumptions carry the decision and what evidence would overturn it, and no one can answer without opening the chat again. The tool did more than help write the analysis. It supplied the first view, the challenge, and the wording of the decision.

This paper takes a position. For work that depends on judgment, AI should be set up so the person forms and defends a view before the tool answers. An answer-first design hands over a finished answer at once. A reasoning-first design asks the person to state a view, name its assumptions, and say what evidence would change it, and only then lets the AI challenge it. That keeps the person's own reasoning in use.

The evidence is thinner than the position. The strongest results come from students. The closest test with working adults is a small experiment with software developers learning a new tool. A field experiment with consultants, a study of medical students, and some surveys and correlations add warning signs. No large workplace trial has tested the two designs on experienced people doing their own expert work, so the position is this paper's inference, not a finding. The paper ends with a one-quarter pilot in one function.

The question is urgent now because AI returns a finished-looking answer in seconds, and many people use it without company training. In a 2024 Microsoft and LinkedIn survey of 31,000 people in 31 markets, 78 percent of AI users brought their own tools to work, while 39 percent had received AI training from their company.[1] The 78 percent counts any personal tool used beside company tools, so it may overstate fully unapproved use. Either way, the habit often forms before the organization sets a standard. The cost shows when the answer is wrong, the case is unusual, or the tool is unavailable.

Each section sets up the next. The student trials show that how people use AI can change what they keep. The workplace evidence shows warning signs but cannot prove the effect. Because proof is missing, the standard is built to be piloted in one function, and the objections come before the pilot.

## What the student trials show

Two randomized trials show how much the way people use AI can matter. They are the paper's best evidence, and both involved students.

### EXHIBIT 1

#### How people use AI can change the result

**17%**

Two separate student trials: answer-first AI left one group 17 percent below a no-AI group on a later exam; a custom AI tutor more than doubled the median learning gain against active learning.

Bastani et al., PNAS 2025; Kestin et al., Scientific Reports 2025 (see Notes).

Bastani and colleagues studied nearly 1,000 high-school students in Turkey. Classes received a standard AI tool, a guarded tutor, or no AI. The standard tool could give a direct answer. The guarded tutor used the same underlying model but led students through hints. On a later exam without AI, the standard-tool group scored 17 percent below the no-AI group. The guarded group was near the no-AI group, though not above it.[2]

Kestin and colleagues studied introductory physics at Harvard. They compared a custom AI tutor with active learning that used peer work and instructor feedback. The AI group had more than twice the median learning gain and used a median of 49 minutes, compared with a 60-minute class. The study covered two topics, a selected student group, and only two weeks. The authors did not claim that the result would carry into complex professional judgment.[3]

The two trials used different students, subjects, and designs, so they are not one comparison. Read together, they support a narrower point: how an AI tool is built can change what students keep. Neither trial tested the first-view step this paper proposes. The guarded tutor led students through hints, which is a neighboring design, and the physics trial compared a tutor with a class, not two AI designs. Neither tested experienced workers, so neither shows the same effect in a company.

## What the workplace evidence adds, and what it cannot

Workplace studies show the same need for care, but none of them settles the question.

The closest to a work setting is a randomized experiment with 52 software developers, mostly professionals or freelancers recruited through a crowd-work platform, who learned a new programming library with or without an AI assistant that could write the code. The AI group scored 17 percent lower on a quiz about the library and was not reliably faster. Developers who asked conceptual questions, or asked for explanations along with code, tended to score higher than those who handed the coding over. They chose how to use the assistant, so that comparison was not randomized. The study had one task, one library, and about an hour of work, and it is a preprint from authors affiliated with Anthropic, an AI developer.[4] It shows what handing work over can cost in learning, not what it does to an expert's judgment in their own field.

In a field experiment with 758 consultants, AI users completed more work, worked faster, and produced higher-rated results on tasks within the model's tested range. On one task outside that range, they were 19 percentage points less likely to reach the correct answer. People whose results fell tended to accept the AI output and question it less.[5]

Persuasive output can make this harder. A 2024 study of four language models from 2023 found them overconfident most of the time.[6] In a study of 257 medical students making 3,855 decisions, AI explanations raised accuracy when the advice was right and lowered it when the advice was wrong, because both kinds sounded convincing. Participants with high competence and appropriate doubt lost the most when the AI was wrong, 12.4 percentage points, while overconfident novices gained. It is an unreviewed preprint, so it warns about persuasive explanations without measuring expert work.[7]

Two more studies add warning signs, with limits. A survey of 319 knowledge workers linked greater confidence in AI with less reported critical-thinking effort. It measured self-reports, not objective performance.[8] A study of 666 people linked frequent AI use with lower critical-thinking scores on a questionnaire, and linked both to cognitive offloading, which means handing mental work to a tool. It was correlational and did not test a better way to use AI.[9]

No large workplace trial has compared answer-first and reasoning-first AI for expert work. No study has directly compared trained workers with self-taught users who bring their own AI tools. Those gaps matter. The operating case in this paper is a bounded inference: when judgment matters, the work should require people to form, test, and own a view before they accept an answer. A workplace trial in which people who wrote a view first did no better than people who asked first would count against it.

## Why the usual responses are not enough

Two common responses leave the gap open. The first is a ban. No source here shows that bans fail, and this paper does not claim they do. Its reading is that when many people already bring their own tools,[1] a ban is more likely to move use out of view than to stop it, and a person using an unapproved tool gets no guidance on when to slow down. That is an inference. It points to a different response: an approved tool that works, with rules for the work where an error is costly.

The second response is a broad “human in the loop” rule. That phrase says nothing about the person’s skill, authority, or task. A reviewer who cannot spot a weak assumption is not a control. A person who sees the work only after the answer is accepted is just a signature step.

In this paper’s reading, four weak points sit under both responses:

- The task is framed only as completion. The worker is rewarded for finishing, not for showing the reasoning.
- The organization measures use and output, but not the quality of the decision.
- The tool removes the productive effort that helps people build skill.
- No named person has to challenge the answer, add missing context, and own the conclusion.

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## Why the judgment is worth protecting

Autor and Thompson offer a labor-economics frame. When technology removes less-skilled tasks from a job, the remaining expert work can become more valuable. When it removes expert work, wages can fall even as employment grows. They do not apply this claim to generative AI. This paper does, as an inference: as AI handles more drafting, search, and routine synthesis, the value of sound judgment can rise. [10]

That judgment includes setting the goal, testing evidence, finding exceptions, and deciding when the model is outside its depth. Faster output has value. It does not replace the need to know whether the output should be trusted, which is why the standard below is aimed at judgment work and not routine work.

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## A better standard: AI as a thinking partner

AI acts as a thinking partner when it helps a person bring out, challenge, and test their own reasoning. The person still owns the conclusion. Use that standard for work with real judgment: strategy, large resource choices, client-facing analysis, and risk decisions. Routine, low-risk work can use more automation. When the model's limits are unknown or the cost of error is high, treat the work as outside the safe range until someone proves otherwise.

### Start with a short pre-mortem

The name comes from Gary Klein's team technique, in which a group imagines that a plan has failed and explains why. The version here is for one person and asks for something different: the assumptions the view depends on, and the evidence that would change it. A lab study with college students found that imagining a hypothetical plan had failed lowered their confidence in it more than a standard critique or a list of pros and cons did. Lower confidence is not the same as better accuracy, and the individual version with AI has not been tested.[11]

Before using AI on high-judgment work, ask the worker to do three things without it:

- State the answer or recommendation they expect, and why.
- Name the two or three assumptions that must be true.
- Say what evidence would change their mind.

Writing the first view should take minutes, not hours, as a planning estimate. If the worker cannot state a view yet, the tool should not supply one. The work waits while the worker gathers facts or asks a colleague, and a record with no first view does not count as reviewed.

Then use AI to challenge that first view. Ask it for the weakest assumption, the best counterargument, and the missing evidence. The worker checks those points against reliable sources and decides what survives. How to check what the tool returns, and what to do with an answer that agrees too easily, is the subject of the companion paper "Confident and Wrong," and this paper does not repeat it.

Related evidence is limited. Ma and colleagues tested a "think the opposite" prompt without AI, and tested confidence calibration in an AI-assisted task. Calibration cut the cases where people ignored good AI advice but not the cases where people followed bad advice.[12] In this paper's reading, the routine changes the worker's posture toward the tool. It does not make the model well calibrated.

## How a reviewer scores the reasoning

Scoring starts from the record the worker leaves: the first view, the assumptions, the evidence that would change it, the tool's challenge, and the final decision. A reviewer who knows the subject, such as a manager or peer, reads the record and checks four things:

- Was the first view written before the tool answered? The time order of the record or chat shows it. If no record exists, as with a personal tool that keeps no history, the check fails and the work moves to the approved tool.
- Are the assumptions specific enough to be wrong? "The market will grow" is too vague to fail. "Renewals stay at today's rate through next year" can be shown wrong.
- Did the worker act on each challenge by checking it against a source, accepting it, or rejecting it with a reason?
- Does the final decision name its owner and say what changed from the first view, or why nothing did?

The reviewer marks each item clear, vague, or missing, and does not decide whether the conclusion is right. A record with a missing item goes back before the decision is made, and a vague item gets a written note. As a planning estimate, reading one record takes about ten minutes. Read every record for the first month of the pilot, then, as a planning assumption, about one in five, with every record for the costliest decisions.

This scoring is the paper's design, and no study has tested it. Its likely failure is that workers write assumptions to satisfy the form, and the second check is the guard. If no one in the function can read a record with knowledge of the subject, that function is the wrong place for the pilot.

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## Two tracks, not one timeline

The standard says what to ask of a person. The two tracks say whom it fits. Some workers still have strong independent judgment and need to protect it. Others may have lost practice after long use of answer-first tools. Those are different problems.

- Track 1 — protect. Install the pre-mortem habit and a real review step over 60–90 days. The goal is to keep current skill active.
- Track 2 — restore. Rebuild independent reasoning through work done without AI, expert feedback, and repeated practice. In this paper's reading, this will take longer.

The 60–90 day range is a planning assumption, not a finding from a validated program. A proven restorative curriculum for experienced workers does not yet exist. Leaders should say so, test carefully, and avoid promising a fast recovery.

Sort people with two inputs. First, learn which tools people use and for which tasks, and make honest answers safe to give. Second, give each person two or three real cases from the function's past decisions, where the outcome is known, and ask without AI for a view, the assumptions, and what would change it. Review the reasoning, not just the answer. As a planning estimate, that takes about an hour per person plus reviewer time. The result decides whether training should protect a skill or help rebuild it. It is not a performance rating.

## The three-layer change model

The standard and the tracks say what to ask of each person. The layers say who must change so the habit lasts: training, the way work is set up, and management review, all at the same time. In this paper's reading, if one layer is missing, deadline pressure is likely to pull people back to answer-first use.

### EXHIBIT 2

#### Support the habit in three places

- 1 **Training design — CHRO and learning team**  
Assess current habits, teach the pre-mortem, explain the tool's limits, and score reasoning rather than output.
- 2 **Work setup — IT, security, and operations**  
Provide a safe tool, require the pre-mortem for high-judgment work, and flag tasks handed fully to AI.
- 3 **Management accountability — managers and executives**  
Ask what changed the worker's view, name who owns the decision, reward sound reasoning, and protect the pilot from a speed-only scorecard.

### Layer 1 — training design

The CHRO and learning team own the training. Frame it as better decision quality, not as tool adoption.

- Assess current habits before teaching one course to everyone.
- Show where the tool works well and where people must slow down.
- Teach people to test assumptions and evidence, not just write prompts.

- Score the reasoning behind a decision, not only the finished output, using the four checks above. The learning team trains the reviewers; a manager or peer who knows the subject does the scoring.

## Layer 2 — work setup

IT, security, and operations own the approved path. Give people a useful, safe option instead of depending on a ban.

- Require the pre-mortem for high-judgment tasks.
- Flag work that people hand fully to AI without an independent check.
- Match the review step to the cost of error and the model's known limits.
- Keep confidential data inside approved tools and access rules.

## Layer 3 — management accountability

Managers and executives make the standard real.

- Ask what the person believed before using AI and what changed their view.
- Name who checked the evidence and who makes the final decision.
- Add reasoning quality to reviews, promotion choices, and project lessons.
- Protect the first pilot from a scorecard that rewards speed alone.

In portfolio and delivery work that I reviewed or advised on, the working rule was that AI proposes and named people check, change, or reject. In one such setup, the tool flagged schedule conflicts, and the owners accepted, changed, or rejected each flag. I saw it as one way to keep an owner on each decision. It is an observation, not evidence that the approach works.

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## Five objections

The pre-mortem slows the work. It does, on the selected work: the time to write a first view, plus a reviewer's time to read the record. Routine work skips it. The cost is worth paying only where an error is expensive, and the pilot should stop if neither rework nor errors fall.

Senior experts will reject it. Some likely will, because a written first view can read as a test of their standing. The record is not a test of the person. It shows where the tool and the expert agree and disagree, and for an expert who already has a view the step is short. The larger risk is a two-tier habit, in which juniors follow the step and seniors

skip it. The executive owner and the most senior reviewers should use the step first, and the pilot should report any skipping as a result.

A written first view may anchor people. It might. A worker who has committed to a view may give too little weight to good AI advice. The Ma study measured confidence calibration, not committing to a view, so it does not settle this. The third step, which asks what evidence would change the view, and the third check, which asks what the worker did with each challenge, are meant to guard against this. The pilot should also track how often the final decision differs from the first view. If it almost never does, read those records first.

Better models will make this unnecessary. Better models may shrink errors on the tasks they handle well. They do not settle who owns the conclusion, and the consultant experiment found errors on a task just outside the model's tested range, which, in this paper's reading, a user may not know they are near.[5] Repeat the pilot's comparison when the model changes, and drop the step for work where it stops paying for itself.

Nothing shows it works for experts. That is true. The evidence is student trials, a small developer experiment, and warning signs. A pilot cannot settle it either. One quarter in one function, without a control group, shows cost, fit, and how reviewers react, not effect. Compare with the function's own earlier quarter or a similar team outside the pilot, and expect both comparisons to be weak. The pilot counts against the position if neither rework nor errors fall, if reviewers cannot tell the records from form-filling, or if experts drop the step when the pilot ends.

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## What to do next

Choose one function where reasoning quality already matters, such as strategy, risk, client advice, or product decisions, and where someone can read a record with knowledge of the subject. Name one executive owner. Before the pilot starts, name who owns each of the three layers. Use the two inputs from the tracks to sort the pilot group, start with the workers who still have strong independent judgment, and report the others separately, because one quarter cannot show restored skill. Define how reviewers will score the reasoning, using the four checks, and how the result will be judged.

For one quarter, require the three-step pre-mortem before people use AI on selected high-judgment work. Record the first view, the assumptions, the evidence that could change it, the AI challenge, and the final human decision. Compare speed, rework (work redone after review), errors (mistakes found after the decision), and review quality with the earlier quarter or the similar team.

At the end of the quarter, decide whether to expand, change, or stop. Do not claim that a short pilot has restored lost skill or proved the effect for experts. It can show whether a reasoning-first design fits the work and whether managers will support it.

The core choice is plain. An answer-first design can make output easier while leaving judgment weaker or hidden. A thinking-partner design makes the worker form a view, test it, and own the conclusion. The model may be the same. The way people work with it is not.

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and has not been independently validated in that specific form.

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### About the author

Marco Policani is an enterprise portfolio, PMO, and AI operating-governance leader. He builds portfolio governance systems where AI carries the analytical load and named humans own every decision — an approach documented across case studies, walkthroughs, and working governance guides on his portfolio site.

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